Hardware Design and Functional Verification Learning Roadmap – Project Plan.

This project plan is a draft of the planned projects, will be updated when each/if each project gets finalised.

1. Sequential Logic and Storage Phase.

The objective is to learn and understand how digital systems store and manipulate information.

a. Basic storage elements

☐SR Latch

☐D Latch

☐D flip-flop

☐D flip-flop with synchronous reset

☐D flip flop with asynchronous reset

☐D flip-flop with enable

☐Parametrized register

☐Register Bank

☐Shift register

b. Counters and sequential generators

☐Binary up-counter

☐Binary down-counter

☐Up/Down counter

☐Modulo-N counter

☐Programmable counter

☐Saturating counter

☐Gray-code counter

☐Johnson counter

☐LSFR

☐Timer

☐Pulse generator

☐Clock divider

c. Memory Components

☐ROM

☐Single-port RAM

☐Synchronous RAM

☐Dual-port RAM

☐Simple dual-port RAM

☐Byte-enable RAM

☐Parametrized RAM

☐Register file

d. Queue and buffer structures

☐Synchronous FIFO

☐Parametrized FIFO

☐Standard-read FIFO

☐Ring buffer

☐LIFO stack

2. Sequential Logic and Storage Phase.

Understand how indepently clocked hardware communicate safely.

a. Syncrhonization components

☐Two flip-flop syncrhonizer

☐Three flip-flop sycnrhonizer

☐Level syncrhonizer

☐Pulse syncrhonizer

☐Toggle syncrhonizer

☐Request/Acknowledge syncrhonizer

b. Asyncrhonous communication structures

☐Asyncrhonous FIFO

☐Asyncrhonous handshake controller

☐Dual-clock memory wrapper

☐CDC control interface

3. Computational Datapaths.

Understand how digital hardware performs arithemtic, logical and data-processing operations.

a. Arithmetic components

☐Half adder

☐Full adder

☐Ripple-carry adder

☐Carry-lookahead adder

☐Adder/Subtractor

☐Incrementer/Decrementer

☐Comparator

☐Singed comparator

b. Logical and bit-manipulation concepts

☐Bitwise logic units

☐Barrel shifter

☐Logical shifter

☐Artithmetic shifter

☐Rotator

☐Priority encoder

☐Leading-zero counter

c. Larger computational units

☐ALU

☐Combinational multiplier

☐Sequential multiplier

☐Pipelined multiplier

☐Restoring divider

☐Non-restoring divider

☐Pipelined divider

☐Fixed-point AU

☐Floating-point AU

d. Error detection and polynomial arithmetic

☐Parity generator/checker

☐CRC generator/checker

☐LSFR based CRC

☐Hamming encoder/decoder

4. Control Logic and Arbitration

Understand how hardware controls and coordinates operations.

a. Control components

☐Moore FSM

☐Mealy FSM

☐Parametrised FSM

☐Protocol controller

☐Command decoder

☐Sequencer

☐Watchdog timer

☐Intrerrupt controller

☐Priority enconder

b. Arbitration components

☐Fixed-priority arbiter

☐Round-Robin arbiter

c. Additional control structures

☐Handshake controller

☐Valid/ready controller

☐DMA controller

☐Intrerrupt prioritizer

☐Timer/Capture controller

5. Communication Protocols and Bus Interfaces

Understand protocols.

a. Serial communication protocols.

☐UART transmitter

☐UART reciever

☐Complete UART controller

☐SPI master

☐SPI slave

☐I2C controller

☐I2C target

☐CAN controller

b. AMBA bus interfaces

☐APB slave

☐APB master

☐APB interconnect

☐AHB-Lite slave

☐AHB-Lite master

c. Buses and additional architectures

☐Wishbone interface

☐APB to AXI bridge

☒SPI to APB bridge

6. Memory Subsystems.

Understand how memory components are organized.

a. Memory Organization

☐Memory address decoder

☐Memory bank controller

☐Banked memory

☐Interleaved memory

☐SRAM controller

☐ROM controller

☐Memory-mapped register block

☐Memory arbiter

b. Cache architectures

☐Direct-mapped cache

☐Fully associative cache

☐Set-associative cache

☐LRU replacement

☐FIFO replacement

☐Random replacement

☐Write-through cache

☐Write-back cache

☐Write buffer

c. Advanced memory structures

☐Cache controller

☐Instruction cache

☐Data cache

☐Multi-level cache

☐TLB

☐Memory ECC controller

7. Processor Architecture

a. Basic processor components

☐Program counter

☐Instruction memory

☐Instruction decoder

☐Register file

☐ALU

☐Immediate generator

☐Branch unit

☐Load/store unit

☐Main control unit

b. Processor implementations

☐Single-cycle processor

☐Multi-cycle processor

☐Five-stage pipelined processor

☐Pipeline register modules

☐Forwarding unit

☐Hazard detection unit

☐Flush controller

☐Branch predictor

c. Advanced processor components

☐CSR unit

☐Exception controller

☐Interrupt controller integration

☐Priviledge-mode controller

☐Instruction cache interface

☐Data cache interface

☐MMU

8. Complete SoC Integration.
    a. System integration components

☐System bus interconnect

☐Crossbar

☐Address decoder

☐Bus bridge

☐Clock generation/distribution

☐Reset distribution

☐Boot ROM

☐SRAM subsystem

☐Interrupt routing

☐DMA integration

b. Peripheral integration

☐UART

☐SPI

☐I2C

☐Timers

☐Interrupt controller

☐GPIO

☐Watchdog

☐DMA

☐Memory controller

☐RAM and ROM

c. Complete system project

☐A processor

☐Memory subsystem

☐Bus interconnect

☐Memory-mapped peripherals

☐Interrupt support

☐Clock and reset architecture

☐Basic software execution environment